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Cellular-connected UAV: Recent Advances and Future Trends*

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***Abstract*—As the application of unmanned aerial vehicles (UAVs) continues to expand, achieving efficient and stable connectivity in integrated air-to-ground communications has become critical. Cellular networks are poised to provide wide-area, high-quality, and secure connections to support UAV operations. However, as the altitude increases, the radio environment changes, posing new challenges for using terrestrial cellular networks to connect UAVs in flight. This paper first introduces the unique air-to-ground channel characteristics of cellular-connected UAVs and explores the impact of these characteristics on communication performance. On this basis, we analyze existing methods for optimizing communication performance and interference mitigation techniques. By comparing the effectiveness of different technical approaches, this paper offers several recommendations for improvement and highlights key directions for future research to further enhance the communication capabilities and reliability of cellular-connected UAVs.**

Keywords—UAVs, Cellular Networks, Integrated Air-to-Ground Communications, Communication Performance

I. Introduction

In recent years, civilian UAVs have been increasingly utilized across various sectors due to their enhanced capabilities and decreasing costs. The application of UAVs is expanding from primarily hobbyist aerial photography to a wide range of tasks in infrastructure, logistics, agriculture, security, and mining, including rescue operations, surveillance, and transportation. Utilizing cellular networks as the access network for UAV users offers advantages such as ubiquitous coverage, seamless mobility management, and highly reliable navigation services, along with robust security and cost-effectiveness [1],[2]. According to relevant reports, the global commercial UAV market is expected to double in size by 2029, with an estimated market value reaching approximately $14 billion. As the number of UAVs and the demand for UAV applications continue to grow exponentially, UAV communication technology faces evolving challenges, necessitating the design of new solutions to support high-performance communication for UAVs [3],[4].

However, as illustrated in Figure 1, existing cellular networks were originally designed with the primary goal of serving terrestrial user terminals [3]. Compared to ground users, UAVs exhibit strong spatial mobility and a higher likelihood of line-of-sight (LoS) channels, which results in base stations being unable to provide highly reliable, high-capacity communication services to UAVs. This creates multiple challenges in the design and deployment of UAV wireless networks.

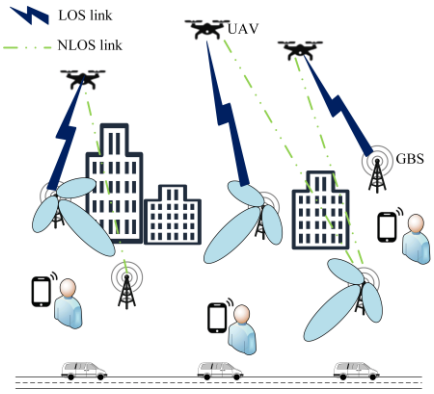

Fig. 1. the three-dimensional coverage situation of ground base stations (GBS).

To achieve seamless coverage for highly mobile UAVs in cellular networks, it is necessary to deeply analyze their three-dimensional mobility and transmission performance. Additionally, the introduction of UAVs into cellular networks leads to more severe air-to-ground interference and UAV-to-base station interference. Addressing these interferences is crucial for ensuring efficient UAV communication.

II. Research on Key Technologies for Cellular-Connected UAVs

With the rapid development of UAV technology, cellular-connected UAVs have become a significant research direction in the field of unmanned aerial vehicles. This technology connects UAVs with ground base station networks, enabling real-time data transmission and remote control, which greatly expands the application scope of UAVs. However, achieving efficient, stable, and secure operation of cellular-connected UAVs still faces many technical challenges. This paper reviews existing research literature to analyze the key technologies that need to be addressed for the development of low-altitude cellular-connected UAVs, providing readers with a clear understanding of the current state of development for

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cellular-connected UAVs. Finally, the paper discusses these issues and presents viewpoints, offering reference value for the future development of 5G base station coverage technology and low-altitude UAVs.

A. Research on A2G Channel Modeling Technology

Traditional channel models are primarily designed for terrestrial wireless communication scenarios, and due to the differing mobility patterns and communication environments of UAVs, these models are significantly limited in their applicability. Compared to ground users, the communication link between a base station and a UAV has a higher probability of being a LoS link. This results in a stronger connection between the UAV and its associated base station, but it also leads to increased inter-cell interference (ICI) [5]. Moreover, modeling the air-to-ground channel for UAVs presents several challenges. First, when a UAV is in flight, the received signal is often influenced by the UAV's body, and factors such as antenna size cannot be ignored. Additionally, when a UAV is flying at high speeds, the effects of Doppler shift and spread on the signal must also be analyzed [6]. These are critical issues that must be considered in UAV air-to-ground channel modeling.

Greenberg et al. employed ray tracing methods to study the statistical characteristics of A2G (air-to-ground) channels in multipath environments [7]. Elnoubi proposed a statistical model for predicting the bit error rate performance of digital modulation techniques, applicable to A2G scenarios [8]. Jiang et al. proposed a 3D elliptical cylinder model suitable for air-to-ground channels, and studied the impact of UAV mobility in the elevation plane and flight altitude on channel characteristics [9].

Taking into account the influence of moving scatterers located at ground terminals, Lian et al. proposed a cylindrical ring-based geometric stochastic channel model for high-altitude platform multiple input multiple output (MIMO) communication systems [10]. Zhu et al. introduced a UAV channel model based on a hemisphere, considering the effects of movement at the transmitter, scatterers, and receiver on channel statistical characteristics, while ignoring static scatterers such as buildings and trees [11]. Currently, few studies consider the combined effects of both moving and static scatterers on the statistical characteristics of UAV communication channels.

B. Research on Communication Performance of Cellular-Connected UAVs

There are still many challenges and issues in providing communication support for UAVs using cellular networks. The signal power transmitted by base stations gradually attenuates during propagation, potentially leading to regions where the signal quality does not meet the basic flight requirements of UAVs. Some progress has been made in this area of research both domestically and internationally. Nguyen et al. studied the downlink communication link from base stations to UAV users, demonstrating the feasibility of using cellular networks to support UAV communications [12]. Insan et al. used MATLAB's Antenna Toolbox to perform simulations and estimates the uplink transmission capacity at various UAV distances and elevations [13]. Ming et al. modeled the downlink interference distribution in a heterogeneous network composed of UAV base stations and cellular base stations, obtaining performance metrics such as coverage and area spectral efficiency. They analyzed the impact of UAV and cellular base station deployment density, cell association strategies, and other factors on network performance [14], although the study did not consider UAVs as users. Azari et al. evaluated the impact of handover overhead and inter-tier association strategies on network performance in heterogeneous cellular scenarios under UAV mobility through simulations [15]. Lyu et al.'s research indicated that using directional antennas on UAVs can effectively reduce cumulative interference experienced by UAVs and improve their coverage [16], but this study did not quantify the impact of interfering base stations on network interference. Du et al. proposed up-tilted antenna model to provide 3D seamless coverage for cellular-connected UAVs, and derives coverage probability for both aerial users (AUs) and terrestrial users (Tus) via stochastic geometry [17]. Wang et al. studied the downlink performance analysis of a UAV in a cellular-connected UAV network with a location-based base station (BS) cooperation strategy [18].

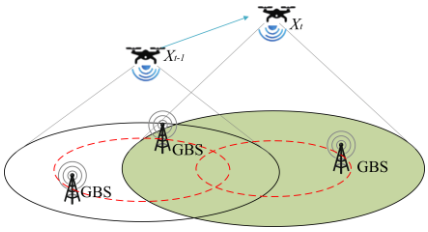

Fig. 2. schematic diagram of drone user mobile switching

Due to the high-speed mobility of UAVs, they typically establish communication links with multiple base stations during flight. When crossing different cells, handover strategies must be considered to optimize network transmission performance, as shown in Figure 2. Existing literature suggests that UAV handover strategies are mostly derived by improving user handover criteria in cellular networks, with consideration of the unique mobility characteristics of UAVs. To address the frequent handover issues caused by the high mobility of UAVs, Sharma et al. proposed a hybrid random mobility model, characterizing handover performance and coverage for ground users in a network composed of UAVs acting as aerial base stations [19]. Fakhreddine et al. studied the relationship between the handover frequency of UAV users and their flight altitude, finding that the handover frequency increases significantly with higher altitudes [20]. Dong et al. introduced a dynamic handover adjustment mechanism that predicts arrival times using flight trajectories to decide whether a handover is necessary, reducing unnecessary handovers [21]. Bai et al. proposed an algorithm that can determine the triggering process of handovers, effectively reducing the number of handovers, handover failures, and the ping-pong effect for UAV users [22]. Kim et al. discussed research trends on utilizing reinforcement learning in cellular networks to enhance UAVs handover performance [23].

C. Research on Interference Control Technology of Cellular-Connected UAVs

Mobile cellular networks are optimized for ground-based broadband communication, and as a result, base station antennas are tilted downward [24] to reduce interference signal strength to neighboring cells. However, as the probability of LoS transmission in the air-ground channel increases, UAVs generate more uplink interference to adjacent cells. If not properly controlled and managed, the

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uplink interference from UAVs can significantly degrade the transmission performance of other co-channel devices.

Understanding the impact of interference on cellular-connected UAVs is one of the key objectives of the 3rd Generation Partnership Project (3GPP) in its research on enhanced LTE support for aerial vehicles [25]. Current research on interference management for cellular-connected UAVs primarily focuses on UAV deployment, cooperation among UAVs, mobility, and trajectory optimization [26]. Efforts have been made to ensure reliable connectivity for cellular-connected UAV users. As wireless networks become increasingly dense, inter-cell interference (ICI) has become a significant issue in NOMA systems. Coordinated Multi-Point (CoMP), a prominent interference mitigation technique, can be applied in UAV communications to address the transmission performance issues faced by UAV users [27]. Amer et al. studied the transmission probability of cellular-connected UAVs and demonstrated that CoMP technology, or optimization of flight altitude, base station height, and antenna tilt angle, can significantly reduce the impact of ICI [28]. The author also proposed a cooperative transmission framework that provides reliable connectivity and mobility support for UAV users [29]. Sun et al. introduced a CoMP-based maximum signal-to-noise ratio (SNR) transmission strategy to maximize the received signal strength of UAV users, showing that the CoMP mechanism can greatly enhance UAV user coverage [30].

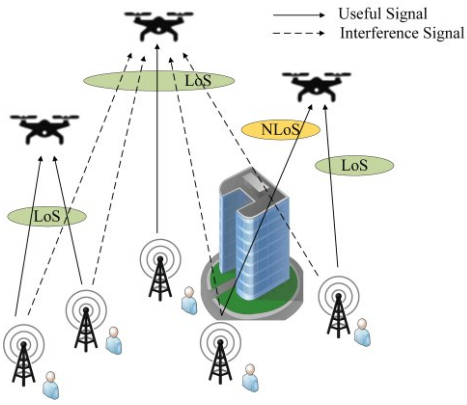

Fig. 3. CoMP-NOMA cellular-connected UAV communication system model

Due to the significant channel differences between base stations and aerial users compared to ground users, NOMA (Non-Orthogonal Multiple Access) technology is highly suitable for scenarios where aerial and ground users coexist [31], as illustrated in Figure 3. Consequently, Sun et al. considered a scenario where UAV users and ground users coexist and proposed a novel interference transmission mechanism based on cooperative beamforming to effectively suppress the interference received by UAV users [32]. To validate the benefits of NOMA in more realistic environments, it is necessary to consider multi-cell networks. In terms of performance analysis, New et al. proposed a stochastic geometry-based performance analysis framework to evaluate the network performance of downlink NOMA in scenarios where aerial and ground users coexist. They derived the user success transmission probability and average achievable rate under different user association strategies [33]. Additionally, Mei et al. introduced an N-NOMA technique for uplink multi-point cooperative transmission, which, compared to traditional NOMA, improves the network service rate and achievable rate [34].

III. Discussion and Analysis

The rapid development of the low-altitude economy makes it essential to achieve three-dimensional signal coverage in the low-altitude airspace. In the early stages of mobile communication network construction, network planning and optimization primarily focused on ground users, leading to large blind spots and significant signal interference in the low-altitude airspace. According to literature research, the existing ground-based station networks struggle to meet the low-altitude coverage requirements, primarily facing the following technical challenges:

1) Lack of dominant coverage cells in low-altitude areas results in coverage blind spots, and increasingly fragmented coverage at higher altitudes;

2) Severe cross-zone coverage issues can cause erroneous handovers, resulting in dropped calls or lag, deteriorated latency, and even communication interruptions;

3) High levels of interference with minimal average difference between the strongest cell level and neighboring cell levels make it challenging to optimize the existing network;

4) Most current base stations are not well-suited for low-altitude coverage, and adjustments to the network significantly impact ground users, affecting network coverage and increase interference between neighboring cells.

Facing the current state of low-altitude networks and their technical challenges, the future development trends of low-altitude communication networks primarily involve several key aspects: establishing more accurate air-to-ground channel models, optimizing the layout of ground stations, and employing power control algorithms and antenna tilt adjustment methods to further enhance low-altitude communication network coverage and address fragmented coverage issues. Additionally, to tackle issues related to cross-zone coverage and interference, implementing more effective cell handover strategies and interference cancellation algorithms can improve network quality and reliability, achieving optimal coverage in low-altitude scenarios. Moreover, artificial intelligence and machine learning can be extensively applied to the management and optimization of low-altitude communication networks. Particularly in areas such as base station handovers, resource allocation, and interference management, AI and machine learning can dynamically adjust network strategies based on UAV flight paths, speeds, and environmental changes. This approach can enhance communication efficiency and ensure optimal utilization of network resources and stability in communication.

IV. Conclusion

In this paper, we systematically review the research progress on cellular-connected UAVs in the areas of channel modeling, communication performance, and interference mitigation technologies. Through comparison of different technical approaches, we find that although current research has made significant progress in enhancing communication efficiency and stability, there are still many challenges in addressing issues such as multipath effects, frequent handovers, and communication stability in high-interference

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environments. Future research should further delve into more intelligent and adaptive air-to-ground integrated network management technologies to comprehensively improve the communication capabilities and security of cellular-connected UAVs, thereby laying a solid foundation for achieving integrated air-to-ground communication networks.

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